

Methylation analysis

Illumina 450K

Toni Gómez

January 2016

Before we get started

- The items for today's workshop have been uploaded to a google drive:
- Part 1:
 - IDAT_files: IDAT methylation files
 - ~~– ChAMP: source code for preprocessing/QC pipeline~~
 - Illustration of preprocessing QC.R
- Part 2:
 - PCA
 - Locus-by-locus analysis
 - Multiple testing
 - Cell mixture

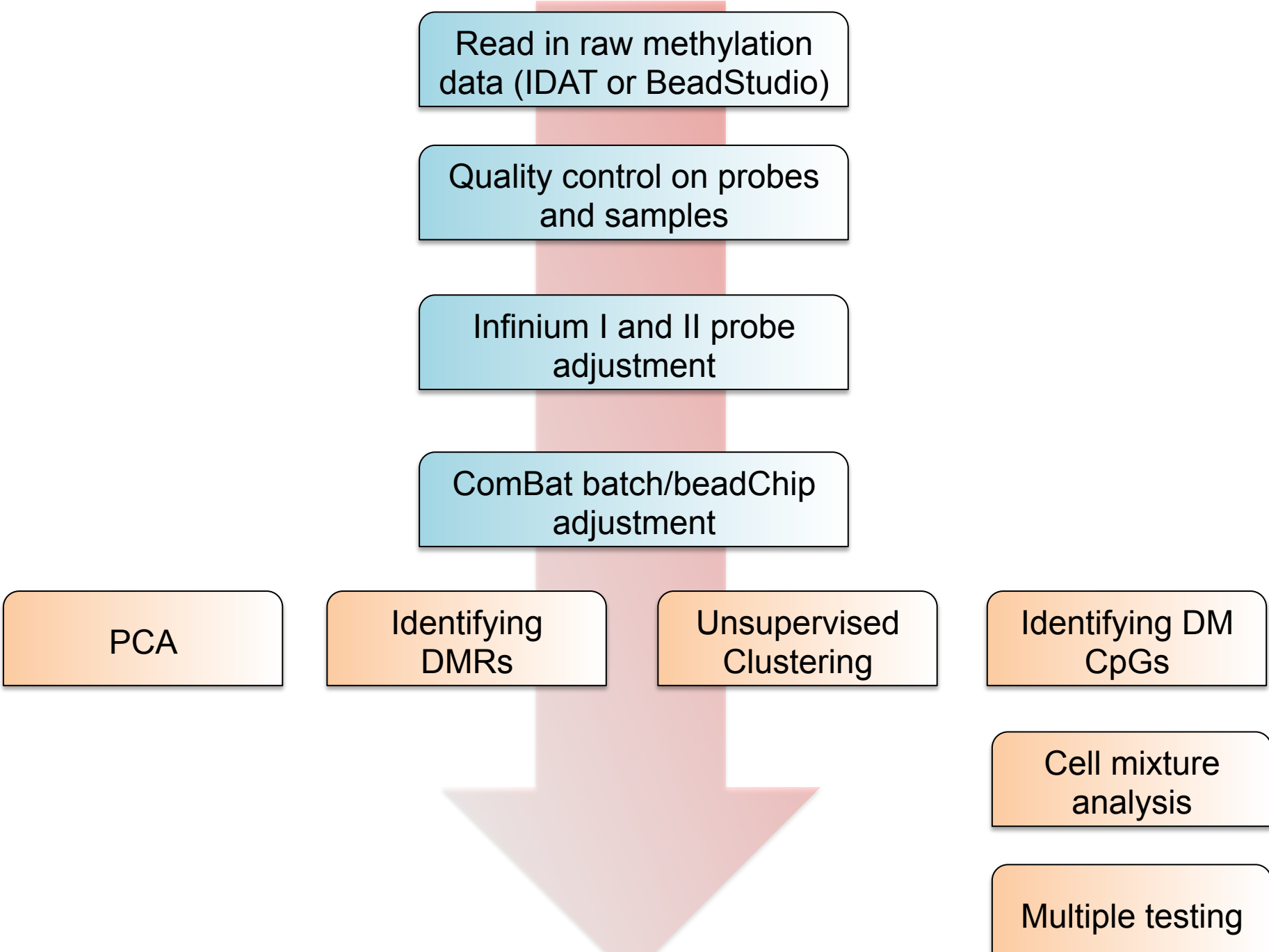
Target audience for this workshop

- Anyone who currently works with or plans to work with array-based methylation data.
- This workshop is NOT:
 - A research seminar
 - A review of DNA methylation or the methods/ techniques used for quantifying it (Bock et al. Nat Reviews 2013)
 - [An R tutorial](#)
- Assumed knowledge:
 - Epigenetics (DNA methylation)
 - Statistics
 - A working knowledge of R

Outline of topics for today's workshop

A proper methylation analysis can be broken down into the following parts:

- Part 1: The preprocessing and quality control of the methylation data
- Part 2: Analysis and inference
- Part 3: Interpretation of the results



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graph TD; A[Read in raw methylation data (IDAT or BeadStudio)] --> B[Quality control on probes and samples]; B --> C[Infinium I and II probe adjustment]; C --> D[ComBat batch/beadChip adjustment]; D --> E[PCA]; D --> F[Identifying DMRs]; D --> G[Unsupervised Clustering]; D --> H[Identifying DM CpGs]; D --> I[Cell mixture analysis]; D --> J[Multiple testing];
```

Read in raw methylation data (IDAT or BeadStudio)

Quality control on probes and samples

Infinium I and II probe adjustment

ComBat batch/beadChip adjustment

PCA

Identifying DMRs

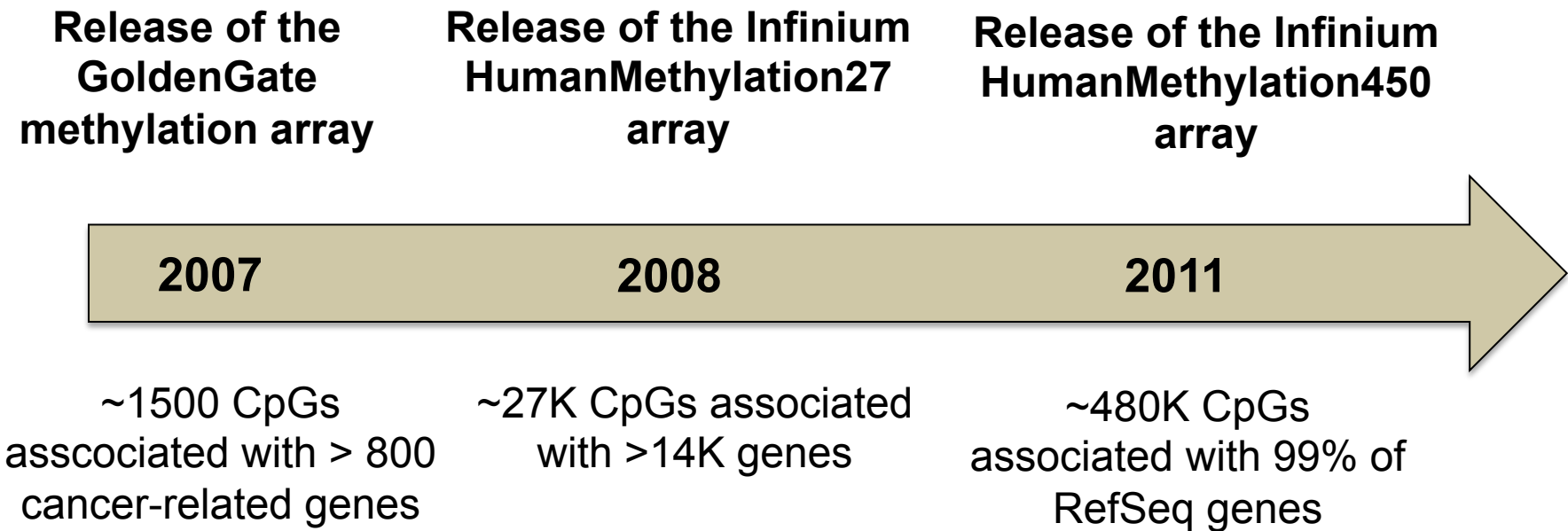
Unsupervised Clustering

Identifying DM CpGs

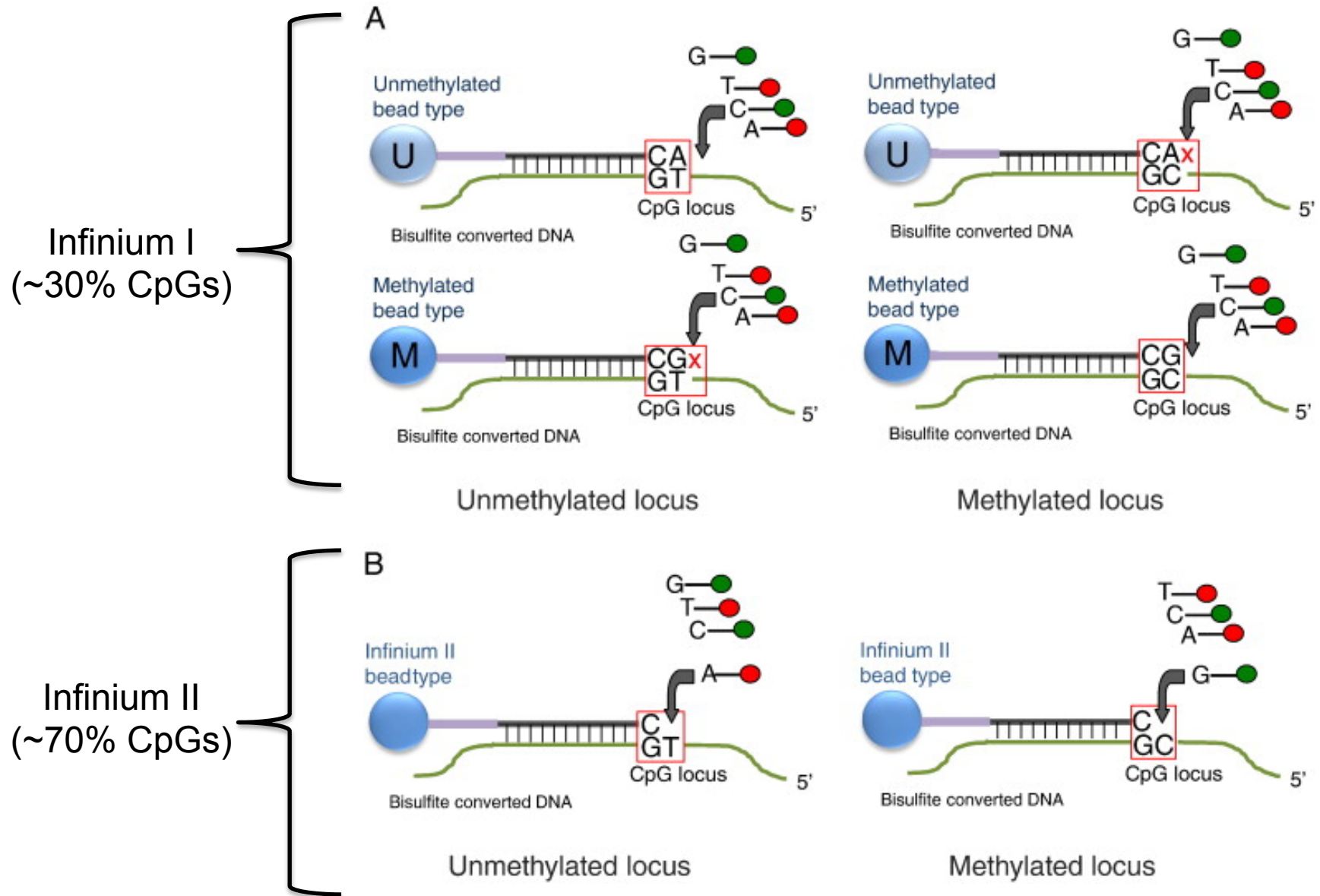
Cell mixture analysis

Multiple testing

A brief history of Illumina high-throughput methylation arrays



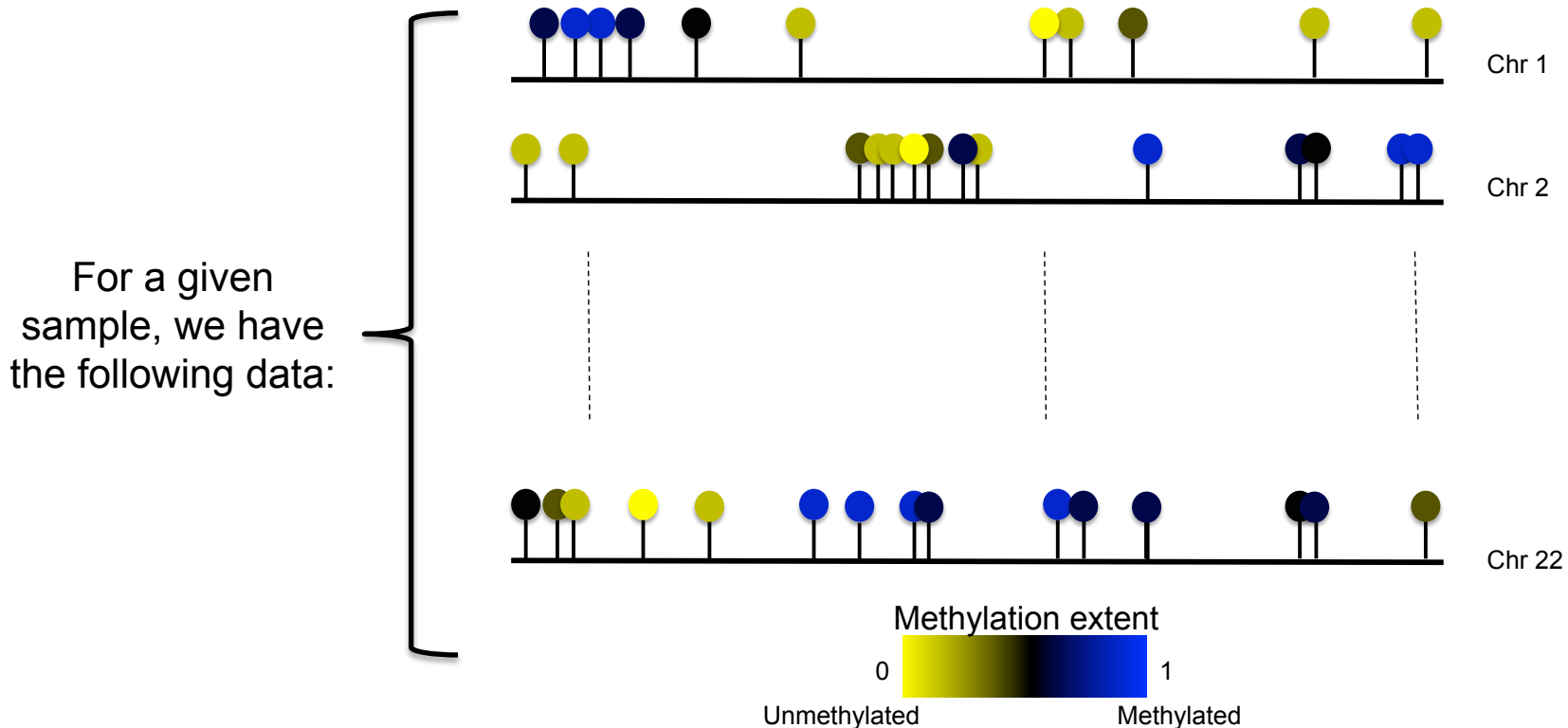
Infinium I and II probe types on the 450K array



Quantifying DNA methylation of a particular CpG site

The beta-value is defined as: $\beta = \frac{M}{M + U + \alpha}$, $0 \leq \beta \leq 1$

- M and U are the methylated and unmethylated signal intensities
- α is an offset (usually 100) to stabilize the beta-values



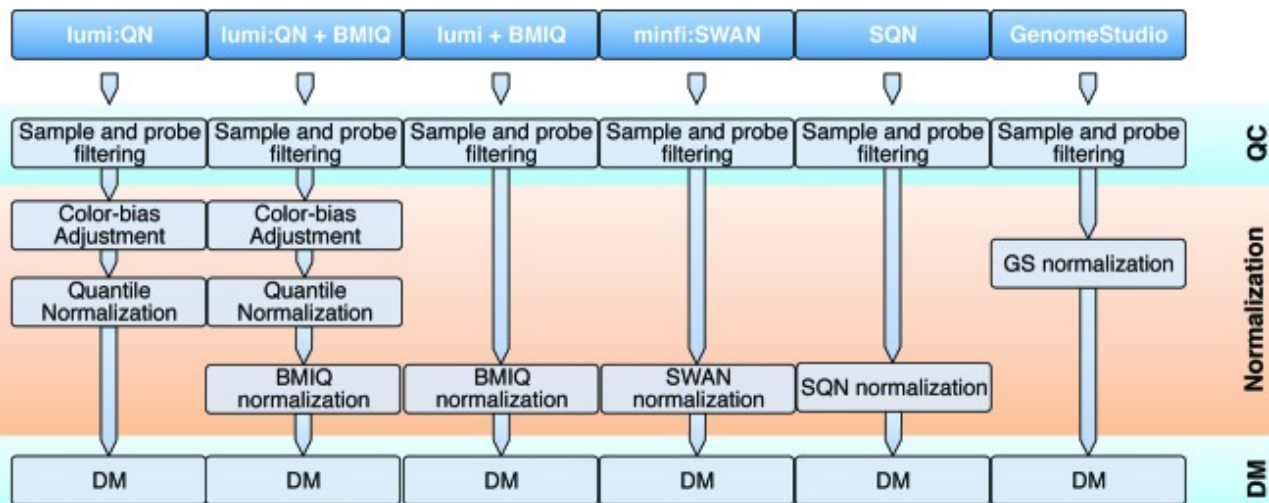
Goals of quality control and preprocessing

- Eliminating or reducing the variability introduced during the experimental process
- Maintaining the biological variation between conditions (i.e., cases/controls)

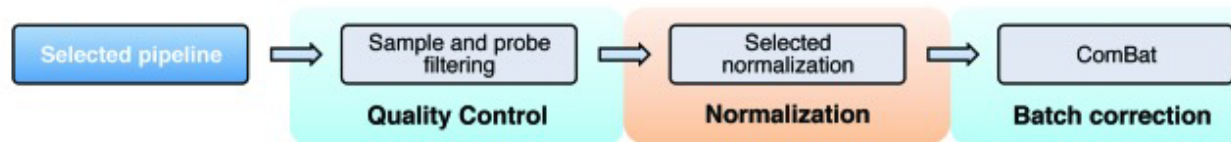
R/BioC Package	File type(s)	Citation/Author(s)
minfi	IDAT	Kasper Daniel Hansen and Martin Aryee
lumi	BeadStudio	Du et al. <i>Bioinformatics</i> (2008)
methyllumi	IDAT or BeadStudio	Davis et al. (2013)
IMA	BeadStudio	Wang et al. <i>Bioinformatics</i> (2012)
ChAMP	IDAT	Morris and colleagues
watermelon	IDAT or BeadStudio	Pidsley et al. <i>BMC Genomics</i> (2013)

For a more comprehensive coverage of publicly available tools for the preprocessing of Illumina methylation data see Bock et al. *Nat Rev* (2012)

Dataset A



Dataset B



Step 1:

Read in raw methylation
data (IDAT or BeadStudio)

Quality control on probes
and samples

Infinium I and II probe
adjustment

ComBat batch/beadChip
adjustment

PCA

Identifying
DMRs

Unsupervised
Clustering

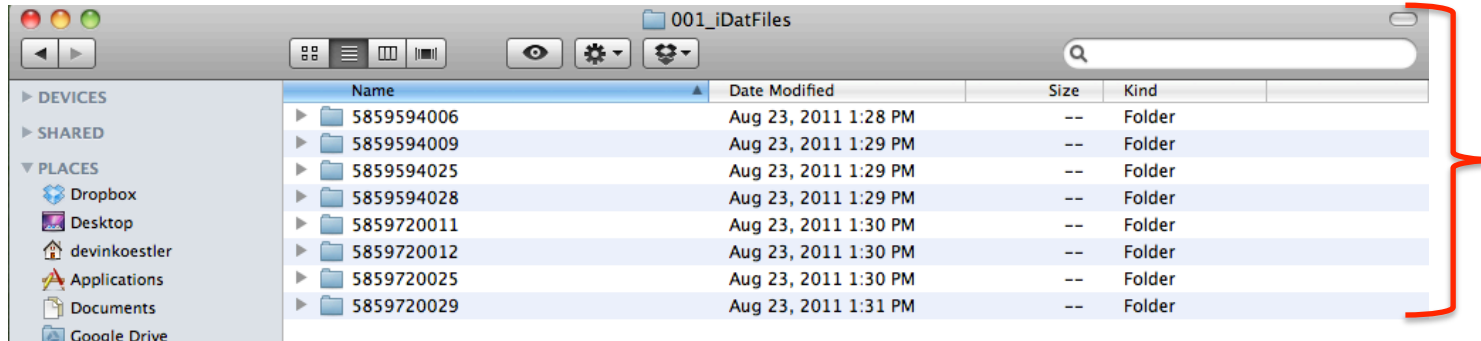
Identifying DM
CpGs

Cell mixture
analysis

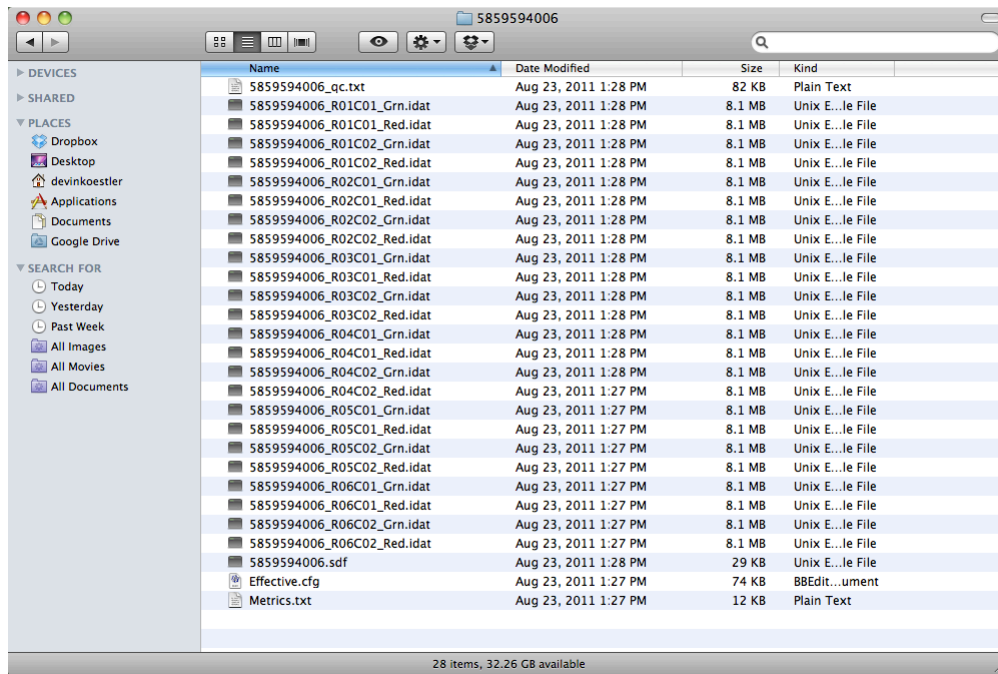
Multiple testing

IDAT Files

IDAT files are typically contained in folders whose name is the chip ID



Folders containing the IDAT files for a single plate (i.e., $n = 96$ samples)



- Each folder represents a different chip, aka slide.
- Red/Green signal intensities for each sample

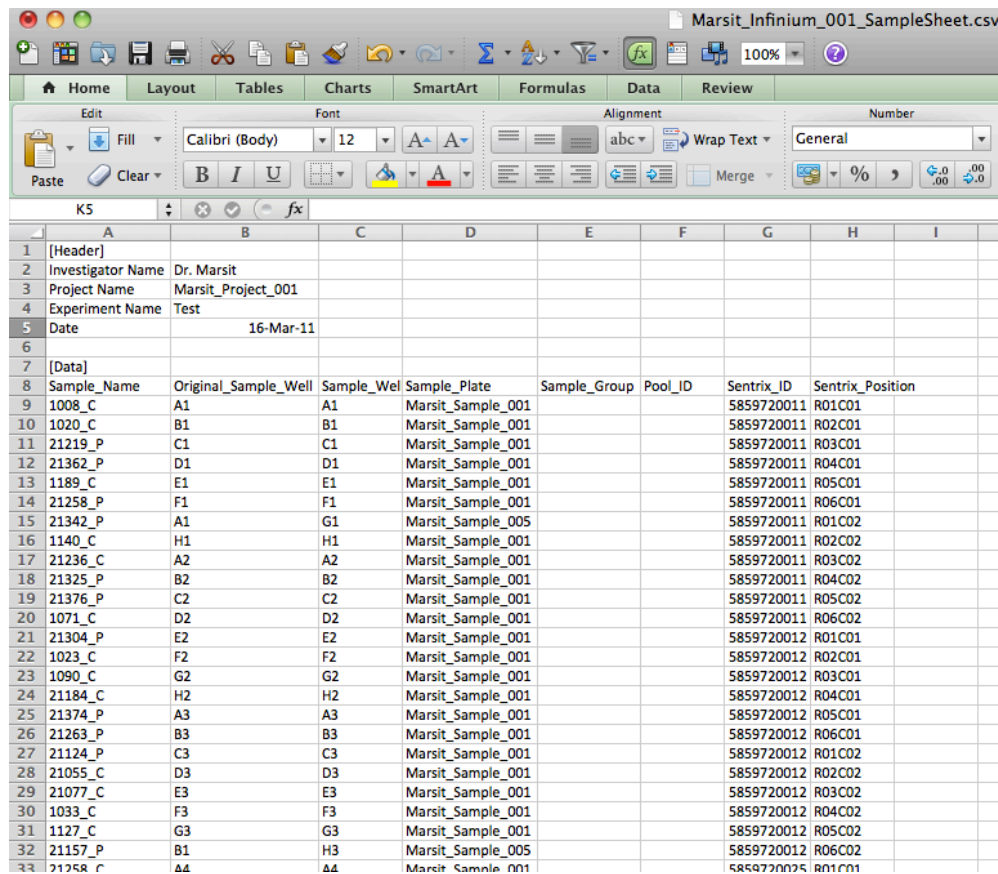
5859594006_R01C01_Grn.idat

Chip/slide Position Chanel

Sample information file

IDAT files are also accompanied with a .txt or .csv file, called the “Sample Information File” or “Sample Sheet”.

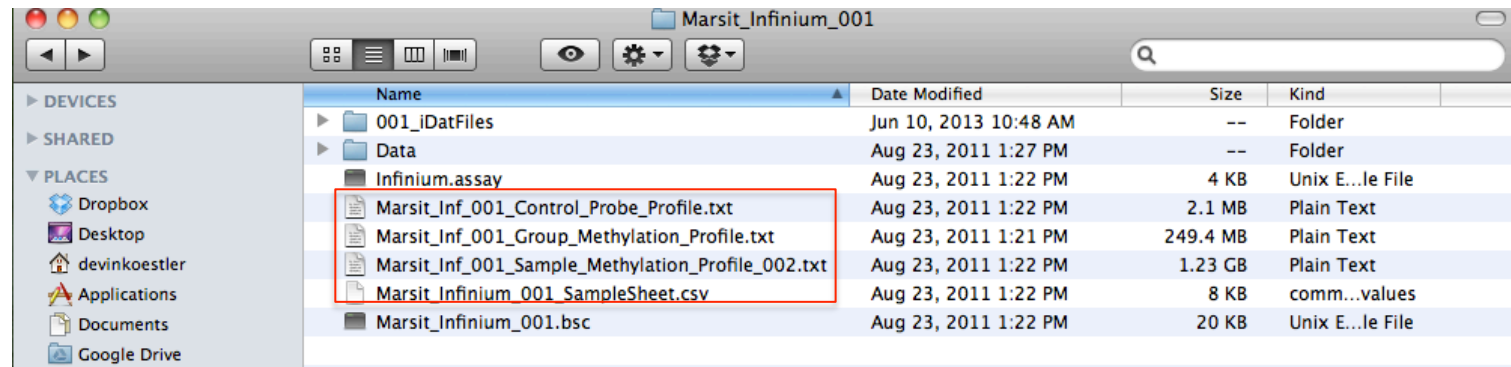
- Provides further details about the samples that were profiled for methylation



Marsit_Infinium_001_SampleSheet.csv							
[Header]							
Investigator Name	Dr. Marsit						
Project Name	Marsit_Project_001						
Experiment Name	Test						
Date	16-Mar-11						
[Data]							
Sample_Name	Original_Sample_Well	Sample_Well	Sample_Plate	Sample_Group	Pool_ID	Sentrix_ID	Sentrix_Position
1008_C	A1	A1	Marsit_Sample_001			5859720011	R01C01
1020_C	B1	B1	Marsit_Sample_001			5859720011	R02C01
21219_P	C1	C1	Marsit_Sample_001			5859720011	R03C01
21362_P	D1	D1	Marsit_Sample_001			5859720011	R04C01
1189_C	E1	E1	Marsit_Sample_001			5859720011	R05C01
21258_P	F1	F1	Marsit_Sample_001			5859720011	R06C01
21342_P	A1	G1	Marsit_Sample_005			5859720011	R01C02
1140_C	H1	H1	Marsit_Sample_001			5859720011	R02C02
21236_C	A2	A2	Marsit_Sample_001			5859720011	R03C02
21325_P	B2	B2	Marsit_Sample_001			5859720011	R04C02
21376_P	C2	C2	Marsit_Sample_001			5859720011	R05C02
1071_C	D2	D2	Marsit_Sample_001			5859720011	R06C02
21304_P	E2	E2	Marsit_Sample_001			5859720012	R01C01
1023_C	F2	F2	Marsit_Sample_001			5859720012	R02C01
1090_C	G2	G2	Marsit_Sample_001			5859720012	R03C01
21184_C	H2	H2	Marsit_Sample_001			5859720012	R04C01
21374_P	A3	A3	Marsit_Sample_001			5859720012	R05C01
21263_P	B3	B3	Marsit_Sample_001			5859720012	R06C01
21124_P	C3	C3	Marsit_Sample_001			5859720012	R01C02
21055_C	D3	D3	Marsit_Sample_001			5859720012	R02C02
21077_C	E3	E3	Marsit_Sample_001			5859720012	R03C02
1033_C	F3	F3	Marsit_Sample_001			5859720012	R04C02
1127_C	G3	G3	Marsit_Sample_001			5859720012	R05C02
21157_P	B1	H3	Marsit_Sample_005			5859720012	R06C02
21158_C	A4	A4	Marsit_Sample_001			5859720012	R01C01

BeadStudio/GenomeStudio files

- Typically a collection of .txt or .csv files that are output from BeadStudio/GenomeStudio
 - Control Probe Profile
 - Group Methylation profile
 - Sample Methylation profile
 - Sample Sheet
- Most likely what you will encounter among publicly available methylation data



Step 1.5:

Read in raw methylation
data (IDAT or BeadStudio)

Quality control on probes
and samples

Infinium I and II probe
adjustment

ComBat batch/beadChip
adjustment

PCA

Identifying
DMRs

Unsupervised
Clustering

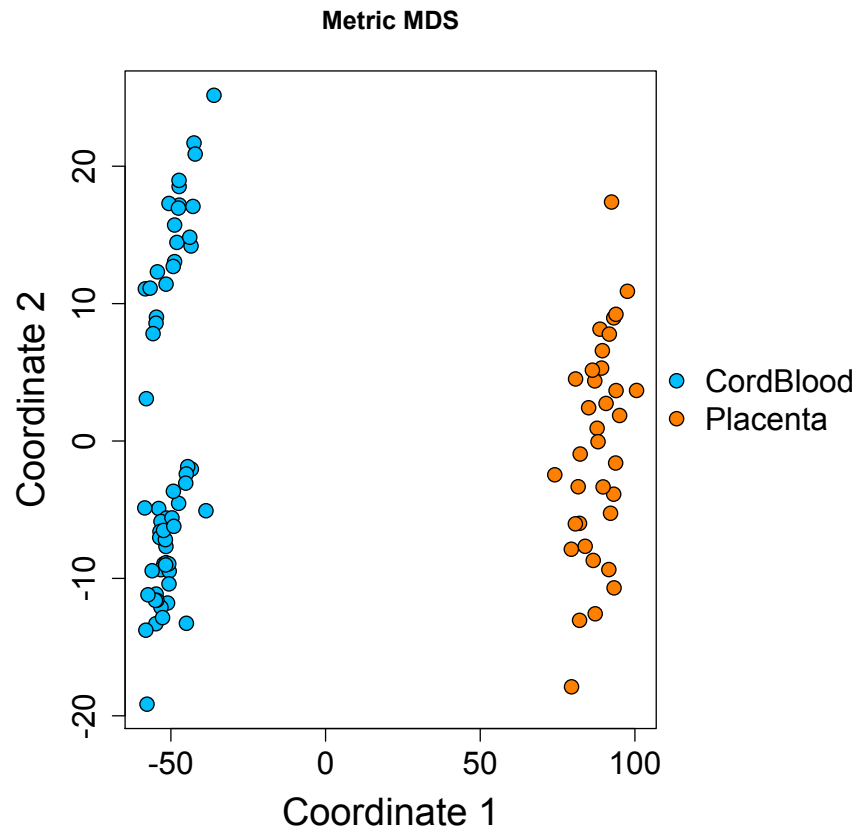
Identifying DM
CpGs

Cell mixture
analysis

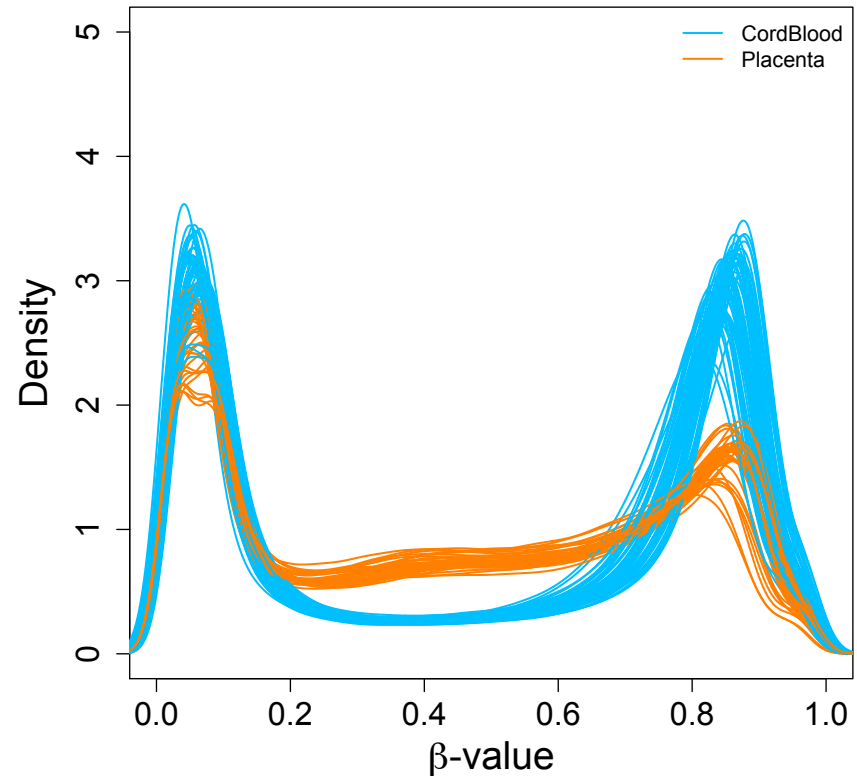
Multiple testing

Step 1.5: Quality control on samples and probes

multidimensional scaling
plot to identify outlying
samples

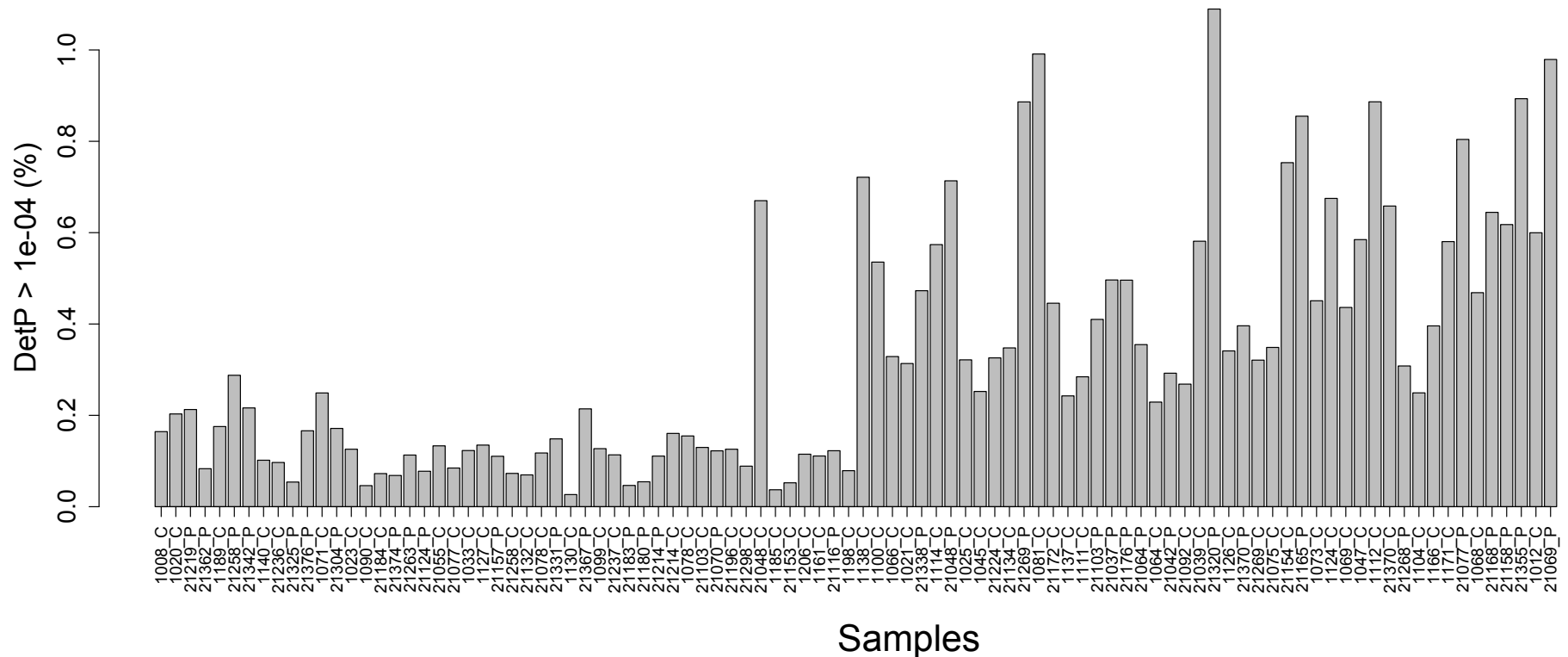


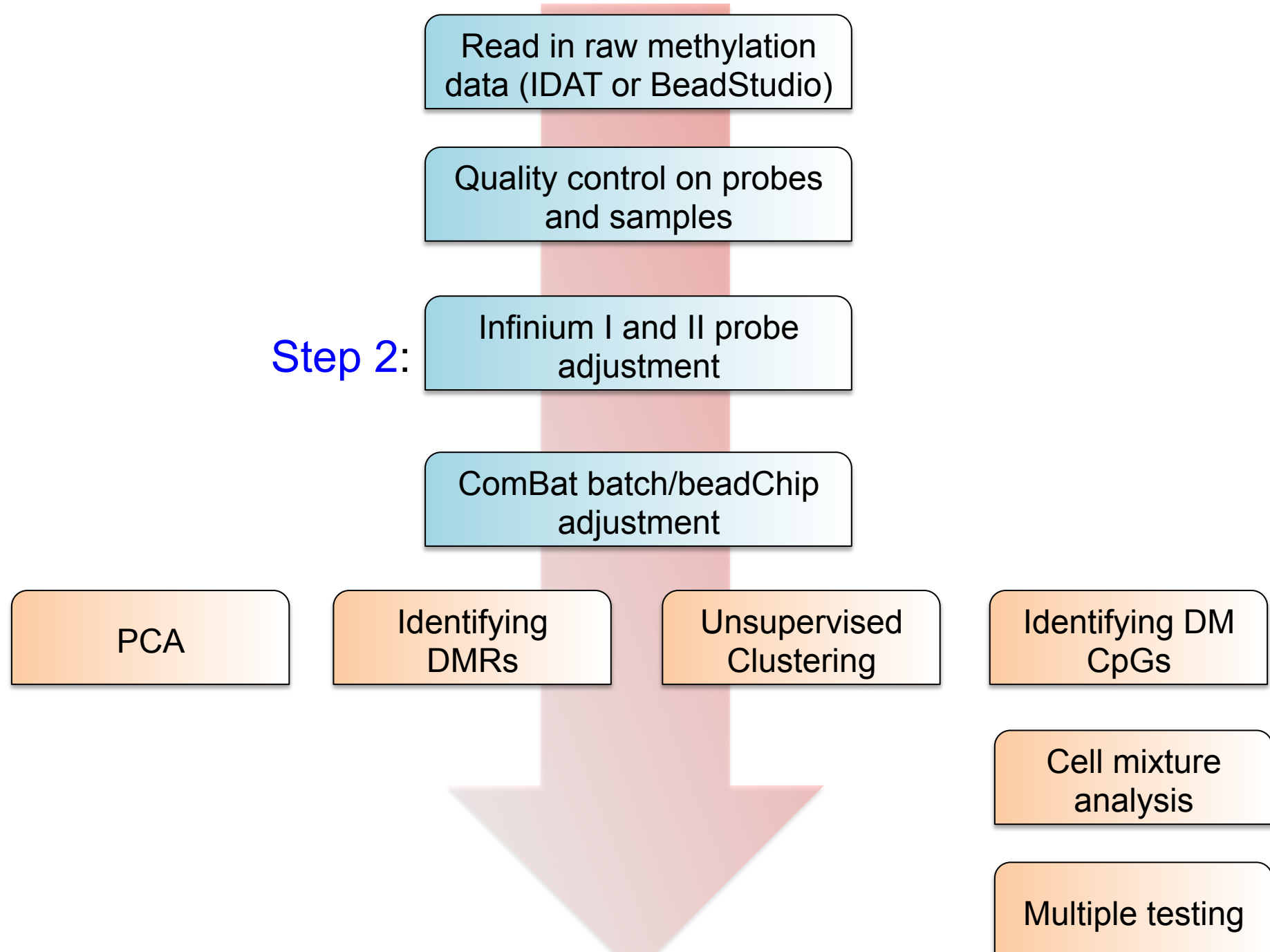
density plot of beta values



Step 1.5: Quality control on samples and probes

barplot of the proportion of detection p-values above some threshold (i.e., > 0.0001)





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Read in raw methylation data (IDAT or BeadStudio)

Quality control on probes and samples

Step 2:

Infinium I and II probe adjustment

ComBat batch/beadChip adjustment

PCA

Identifying DMRs

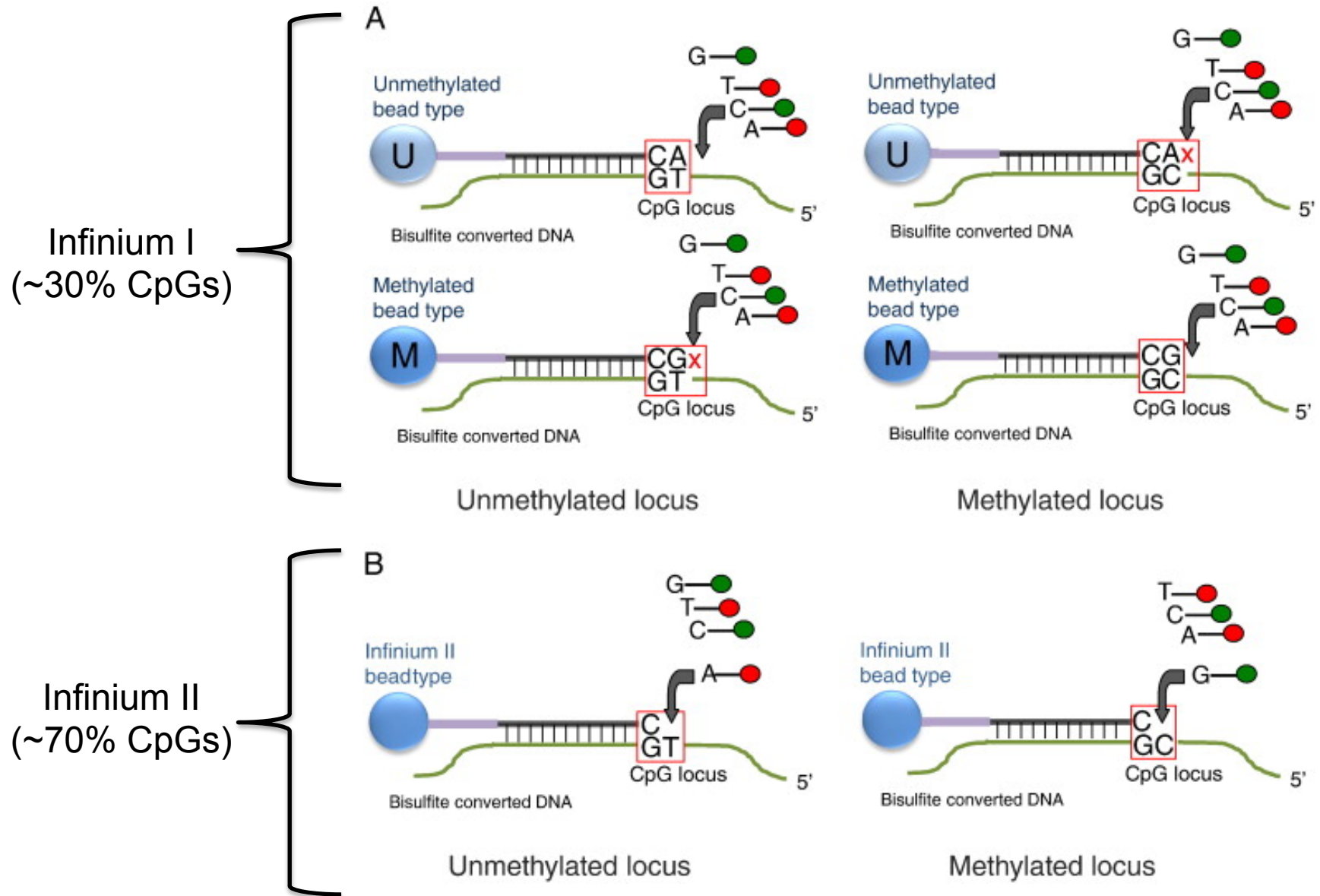
Unsupervised Clustering

Identifying DM CpGs

Cell mixture analysis

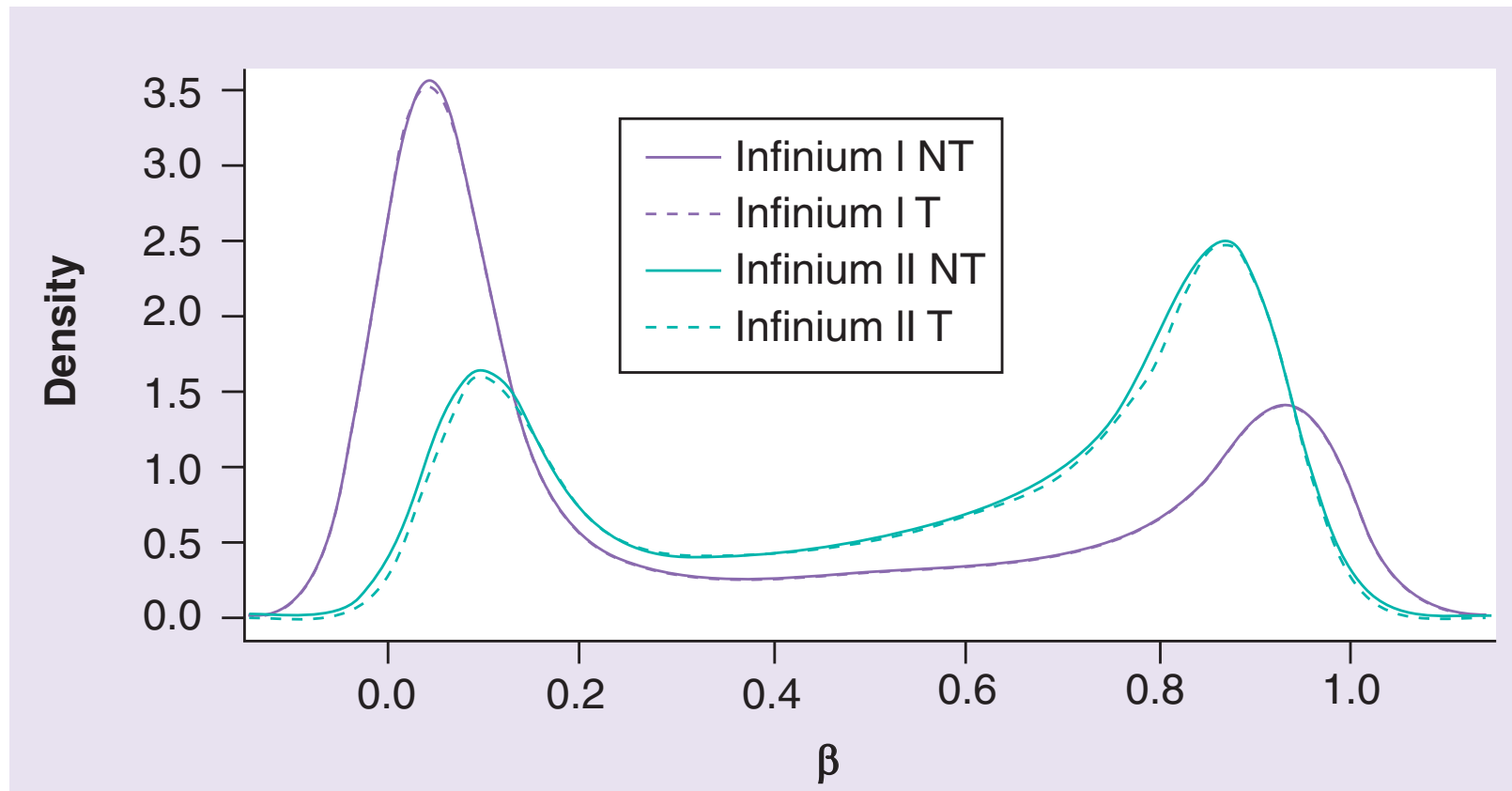
Multiple testing

Infinium I and II probe types on the 450K array



Distribution of Infinium I and II probe types

A shift in the distribution of beta-values based on probe type was first reported in Dedeurwaerder et al. (2011)



Distribution of Infinium I and II probe types

Making the distributions of Infinium I and II probes comparable is important for several reasons:

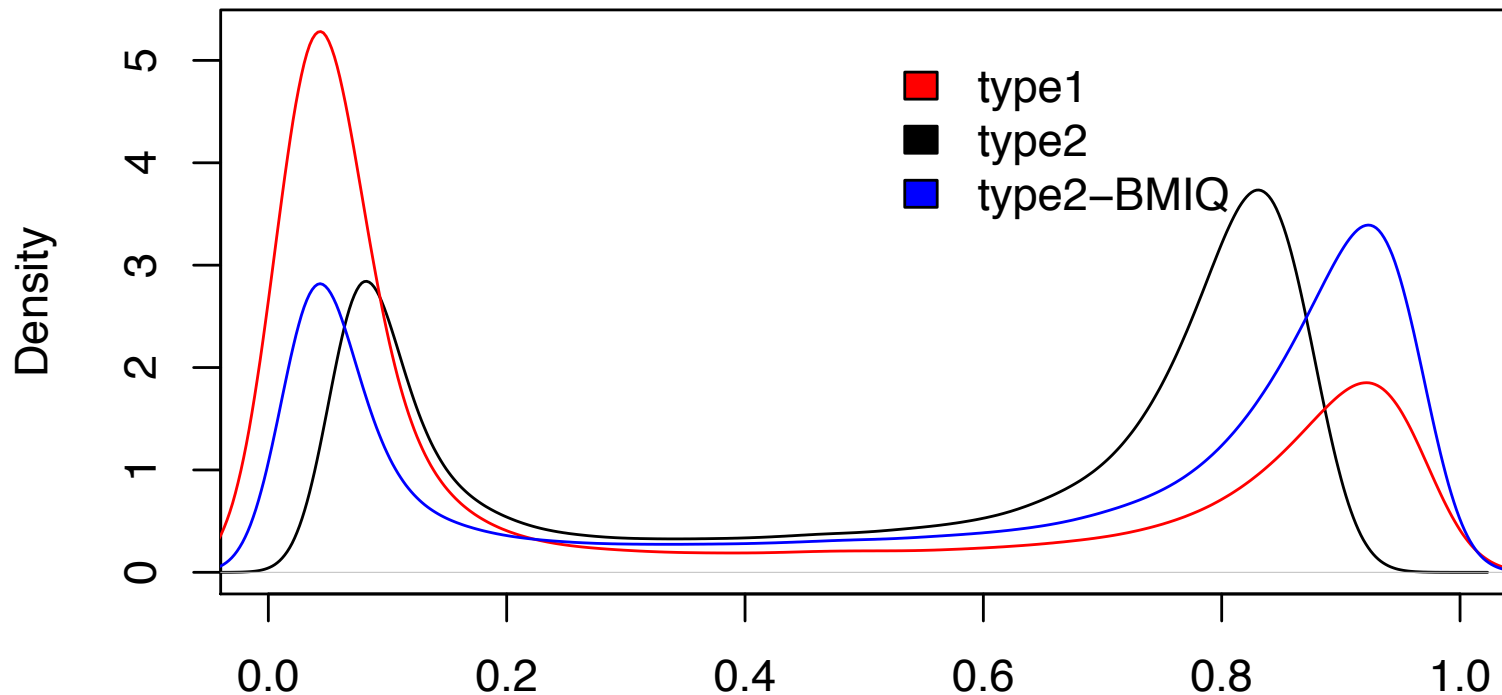
- Enrichment bias toward Infinium 1 probes when ranking probes in a supervised analysis
- Methods for identifying differentially methylated regions (Jaffe et al. 2012) assume that these regions are comparable
- Unsupervised dimension reduction algorithms (Houseman et al. 2008 and Koestler et al. 2010)

Several methods have been proposed:

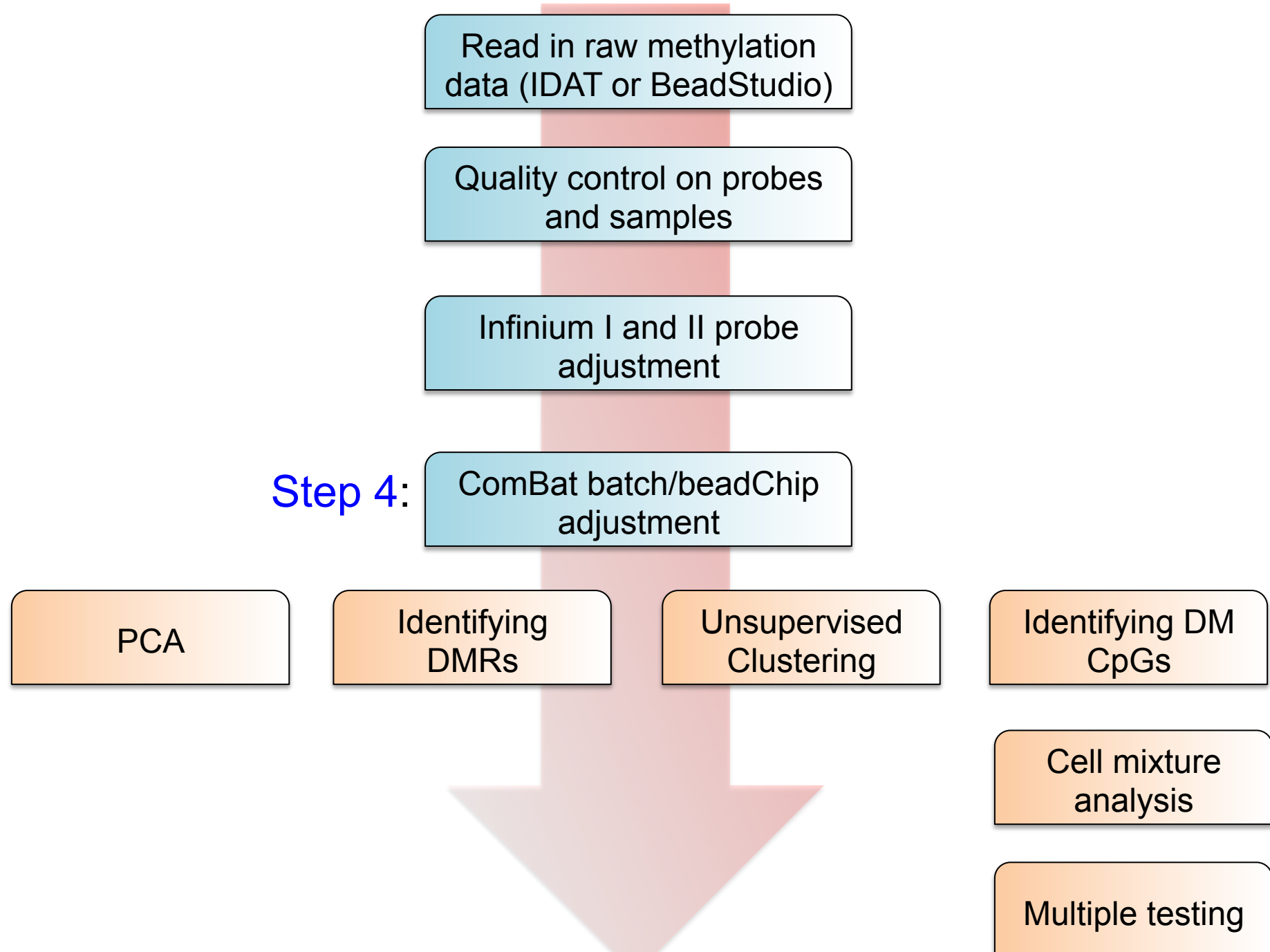
- Peak-based correction (PBC) (Dedeurwaerder et al 2011)
- Subset Quantile Normalization (SQN) (Touleimat and Tost 2012)
- Subset Quantile within array Normalization (SWAN) (Maksimovic et al. 2012)
- **Beta Mixture Quantile dilation (BMIQ)** (Teschendorff et al. 2013)

Step 2: Infinium I and II probe adjustment

- If plotsBMIQ = T, this function will also generate plots depicting the BMIQ adjustment for each of the study samples.
- Located in a folder called resultsChamp, within the IDAT_files folder.



N = 340967 Bandwidth = 0.022



```
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Read in raw methylation data (IDAT or BeadStudio)

Quality control on probes and samples

Infinium I and II probe adjustment

Step 4:

ComBat batch/beadChip adjustment

PCA

Identifying DMRs

Unsupervised Clustering

Identifying DM CpGs

Cell mixture analysis

Multiple testing

Step 5: Removal of cross-hybridizing probes?

RESEARCH PAPER

Epigenetics 8:2, 203–209; February 2013; © 2013 Landes Bioscience

Discovery of cross-reactive probes and polymorphic CpGs in the Illumina Infinium HumanMethylation450 microarray

Yi-an Chen,^{1,2,†} Mathieu Lemire,^{3,†} Sanaa Choufani,¹ Darci T. Butcher,¹ Daria Grafodatskaya,¹ Brent W. Zanke,^{3,4} Steven Gallinger,⁵ Thomas J. Hudson^{3,6} and Rosanna Weksberg^{1,2,7,8,*}

- A list of cross-hybridizing probes are given in the Supplementary Materials for the above manuscript.
- Remove prior to analysis or check *a posteriori*?